Publication List Leander Thiele
June 4, 2026

L. Thiele, *Machine Learning Techniques for Astrophysics and Cosmology: Simulation-Based Inference*, 2026, `arXiv:2605.10719 [astro-ph.CO]`, Invited chapter for the edited book "Machine Learning Techniques for Astrophysics and Cosmology" (Eds. C. Bambi, V. Kashyap, S. Shashank, N. Yoshida, Springer Singapore, expected in 2026)

A. Hell, **L. Thiele**, *LLMs with in-context learning for Algorithmic Theoretical Physics*, 2026, `arXiv:2605.08212 [cs.LG]`, poster at ICML 2026 AI4Physics workshop

L. Thiele, *Bayesian Cosmic Void Finding with Graph Flows*, 2026, `arXiv:2602.14630 [astro-ph.CO]`, OJAp 9, poster at ICML 2026 AI4Physics workshop

J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026, `arXiv:2601.20669 [astro-ph.CO]`

G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the y-distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025, `arXiv:2512.03038 [astro-ph.CO]`

A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025, `arXiv:2511.20423 [astro-ph.CO]`

B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025, `arXiv:2510.08573 [astro-ph.CO]`, poster at ML4PS workshop NeurIPS 2025

B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025, `arXiv:2507.20378 [cs.LG]`, spotlight talk at ML4Astro workshop (colocated with ICML 2025)

J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025, `arXiv:2507.12315 [astro-ph.CO]`

L. Thiele, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025, `arXiv:2507.00514 [astro-ph.CO]`, poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A. LaPosta, et al (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025, JCAP 11, 063, `arXiv:2503.14454 [astro-ph.CO]`

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024, poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

L. Thiele, *De-baryonifying halos via optimal transport*, 2024, `arXiv:2411.18399 [astro-ph.CO]`

J. Armijo, G.A. Marques, C.P. Novaes, **L. Thiele**, J.A. Cowell, D. Grandón, M. Shirasaki, J. Liu, *Cosmological constraints using Minkowski functionals from the first year data of the Hyper Suprime-Cam*, 2025, MNRAS 537, 4, arXiv:2410.00401 [astro-ph.CO]

C.P. Novaes, **L. Thiele**, J. Armijo, S. Cheng, J.A. Cowell, G.A. Marques, E.G.M. Ferreira, M. Shirasaki, K. Osato, J. Liu, *Cosmology from HSC Y1 Weak Lensing with Combined Higher-Order Statistics and Simulation-based Inference*, 2025, PRD 111, 8, arXiv:2409.01301 [astro-ph.CO]

A. Kogut, E. Switzer, D. Fixsen, N. Aghanim, J. Chluba, D. Chuss, J. Delabrouille, C. Dvorkin, B. Hensley, J.C. Hill, B. Maffei, A. Pullen, A. Rotti, A. Sabyr, **L. Thiele**, E. Wollack, I. Zelko, *The Primordial Inflation Explorer (PIXIE): Mission Design and Science Goals*, 2025, JCAP 004, 020, arXiv:2405.20403 [astro-ph.CO]

S. Cheng, G.A. Marques, D. Grandón, **L. Thiele**, M. Shirasaki, B. Ménard, J. Liu, *Cosmological constraints from weak lensing scattering transform using HSC Y1 data*, 2025, JCAP 01, 006, arXiv:2404.16085 [astro-ph.CO]

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D. Wadekar, **L. Thiele**, J.C. Hill, S. Pandey, F. Villaescusa-Navarro, D.N. Spergel, M. Cranmer, D. Nagai, D. Anglés-Alcázar, S. Ho, L. Hernquist, *The SZ flux-mass (Y-M) relation at low halo masses: improvements with symbolic regression and strong constraints on baryonic feedback*, 2022, MNRAS 522, 2, arXiv:2209.02075 [astro-ph.CO]

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